



STUDENT NUMBER

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MATHEMATICS

EXTENSION 1

HSC ASSESSMENT TASK 2

15th March, 2012

TIME ALLOWED: *Reading time: 5 minutes*
 Working time: 2 hours

INSTRUCTIONS: This paper consists of 10 multiple choices and 4 fifteen mark questions
 ANSWER all questions
 Start each question in a new booklet
 Marks may not be given for untidy and poorly arranged work.
 All relevant working should be shown.

THIS QUESTION PAPER MUST NOT BE REMOVED FROM THE EXAMINATION ROOM

OUTCOMES ASSESSED:

H1 - seeks to apply mathematical techniques to problems in a wide range of practical contexts H2 - constructs arguments to prove and justify results H4 - expresses practical problems in mathematical terms based on simple given models H5 - applies appropriate techniques from the study of calculus and trigonometry to solve problems H6 - uses the derivative to determine the features of the graph of a function H7 - uses the features of a graph to deduce information about the derivative H9 - communicates using mathematical language, notation, diagrams and graphs Plus Preliminary from P1 to P8 HE1 - Appreciates interrelationships between ideas drawn from different areas of mathematics HE6 - Determines integrals by reduction to a standard form through a given substitution
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This assessment task constitutes 25% of the Higher School Certificate Course Assessment.

Multiple choices (1 mark each)
Use the answer sheet for Q1 to 10

1. If $f(x) = \begin{cases} 8x^3 & \text{if } x > 3 \\ 3x^2 - 2 & \text{if } 0 \leq x \leq 3 \\ 9 & \text{if } x < 0 \end{cases}$ then $f(3) + f(1) + f(-1) = \dots\dots$

(A) 35

(B) 226

(C) 233

(D) 53

2. $\lim_{\theta \rightarrow 0} \frac{\tan \frac{\theta}{6}}{\theta} = \dots\dots\dots$

(A) 6

(B) $\frac{1}{6}$

(C) 1

(D) 3

3. The coordinate of the focus of the parabola $x^2 - 2x + 2y - 11 = 0$ is:

(A) $(1, 5 \frac{1}{2})$

(B) $(1, 6 \frac{1}{2})$

(C) -1, -6)

(D) (2, 3)

4. If $\tan \theta = \sqrt{3}$ and $\cos \theta < 0$, then $\sin \theta = \dots$

(A) 1

(B) $\frac{1}{2}$

(C) $\frac{\sqrt{3}}{2}$

(D) $-\frac{\sqrt{3}}{2}$

5. Consider the function $f(x) = \frac{x^2 - 1}{x - 4}$,

As $x \rightarrow \infty$, $f(x) \rightarrow ?$

(A) 0

(B) $\frac{1}{4}$

(C) $x - 1$

(D) x

6. Which of the following is true?

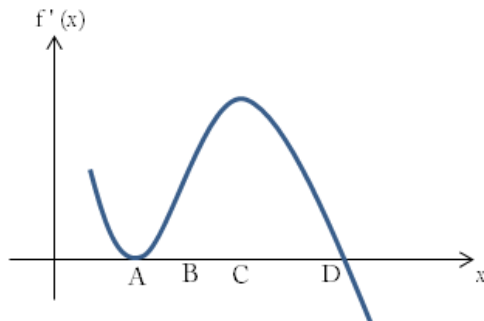
(A) $\log_{10} \left(\frac{x^2}{y} \right) = \frac{2 \log_{10} x}{\log_{10} y}$

(B) $\log_2 16 + \log_2 8 = 12$

(C) $e^{\ln x} = \ln x$

(D) $\log_3 \frac{1}{81} = -4$

7. Below is the graph of $f'(x)$. $f(x)$ has the horizontal point of inflexion at :



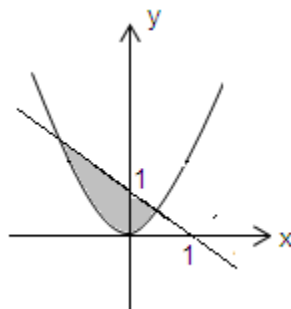
(A) A

(B) B

(C) C

(D) D

8. The inequalities which correspond to the region shaded is:



(A) $y \leq 1 - x$ and $y \leq x^2$

(B) $y < 1 - x$ and $y \geq x^2$

(C) $y \leq 1 - x$ and $y \geq x^2$

(D) $y \leq 1 + x$ and $y \geq x^2$

9. Consider the function $f(x) = \frac{x^2 - 2x - 3}{(x+1)(x-4)}$. Which of the following statement is true?

(A) $f(x)$ has horizontal asymptotes at $x = -1$ and $x = 4$

(B) $f(x)$ has vertical asymptotes at $x = -1$ and $x = 4$

(C) $f(x)$ is discontinuous at $x = -1$

(D) None of the above.

10. The polynomial $P(x) = x^3 + px^2 + qx + r$ has real roots $\sqrt{\beta}$, $-\sqrt{\beta}$ and α . Find the value for r in term of p and q .

(A) $r = -\frac{p}{q}$

(B) $r = \frac{p}{q}$

(C) $r = -pq$

(D) $r = pq$

Total marks – 60

Attempt all Questions

All questions are of equal value

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available

Question 11 (15 marks) Use a SEPARATE writing booklet

Mark

- a) Prove that $\frac{1}{\sqrt{5}+1} + \frac{1}{\sqrt{5}+3}$ is rational. 2
- b) Differentiate $e^{2x}(\cos x - \sin x)$ 2
- c) Solve the inequality $\frac{x-1}{2x-1} \leq 3$ 3
- d) Simplify $\frac{4^{n-1} \times 2^n}{8^{n-3}}$ 2
- e) The acute angle between the lines $y=3x+5$ and $y=mx+4$ is 45° . Find the two possible values of m . 3
- f) i) Sketch the graph of $y=|3-2x|$ 2
- ii) For what values of b does $|3-2x|=b-x$ have two solutions? 1

Question 12 (15 marks) Use a **SEPARATE** writing booklet

Mark

a) Find the indefinite integral: $\int \sqrt{x} \left(1 + \frac{1}{x} \right) dx$ 1

b) Find the exact answer of $\int_{\sqrt{2}}^2 \frac{4x+2}{x^2+x+1} dx$ 2

c) i) Determine whether $y = \sin x \cdot \tan^2 x$ is an odd or even function. 2

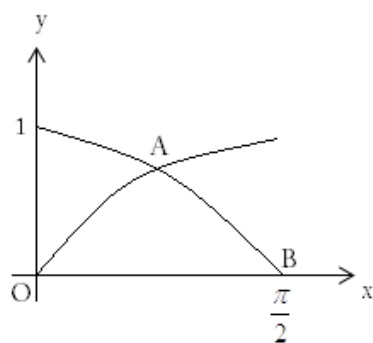
ii) Hence, otherwise evaluate $\int_{-1}^1 \sin x \tan^2 x dx$. 1

d) Using the substitution $u = 2x^2 + 1$ to find $\int x(2x^2 + 1)^{\frac{5}{3}} dx$ 2

e) Show that $\frac{1 + \sin^2 x}{\cos^2 x} = 2 \sec^2 x - 1$. 2

Hence, otherwise evaluate $\int_0^{\frac{\pi}{3}} \frac{1 + \sin^2 x}{\cos^2 x} dx$ 2

f) 3



NOT DRAWN TO SCALE

The diagram shows portions of the graphs $y = \sin x$ and $y = \cos x$. Find the area of the region enclosed by the arc OA, the arc AB and the interval OB.

Question 13 (15 marks) Use a **SEPARATE** writing booklet

Mark

a) Let $f(x) = x + \log_e x$.

(i) Write down the natural domain for $f(x)$. 1

(ii) Show that for all values of x in the natural domain, $y = f(x)$ is increasing 1

(iii) Show that the curve $y = f(x)$ cuts the x axis between $x = 0.5$ and $x = 1$. 1

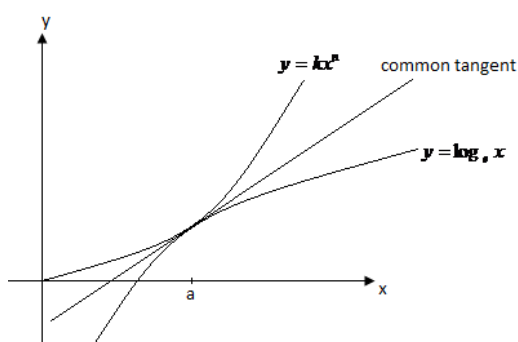
(iv) Use Newton's method with a first approximation of $x = 0.5$ to find a second approximation to the root of $x + \log_e x = 0$. 2

b) Find the exact volume of the solid formed when the curve $y = \log_e (2x)$ is rotated about the y -axis from $y = 1$ to $y = 3$. 3

c) i) Show that $\frac{d}{dx} \ln \left(\frac{x-1}{x+1} \right) = \frac{2}{(x-1)(x+1)}$ 2

ii) Hence find the exact answer of $\int_2^5 \frac{2}{(x-1)(x+1)} dx$ 1

d)



The graphs of the functions $y = kx^n$ and $y = \log_e x$ have a common tangent at $x = a$, as shown in the diagram.

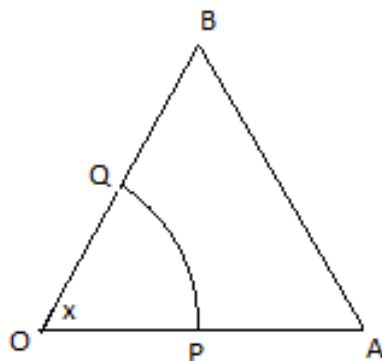
i) Show that $a^n = \frac{1}{nk}$ 2

ii) Express k as a function of n by eliminating a . 2

Question 14 (15 marks) Use a **SEPARATE** writing booklet

Mark

a)

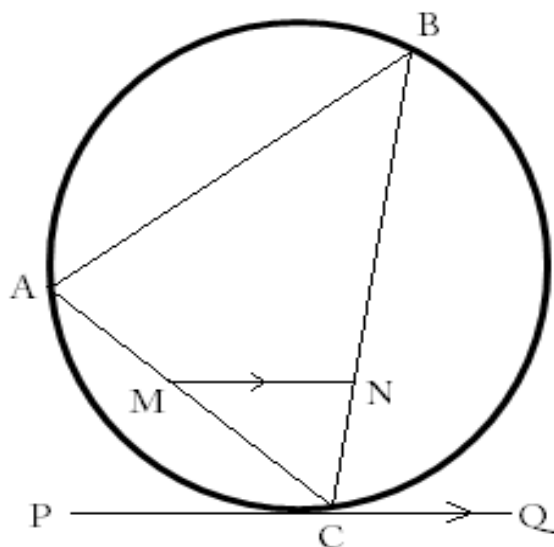


In $\triangle OAB$, $OA = OB = a$, which is a constant, $\angle AOB = x$ radians, where x is variable. PQ is a circular arc, centre O and radius r . If the area of $\triangle OAB$ is twice that of sector OPQ,

- i) Express r^2 in terms of a and x 2
 - ii) Find r when $\angle AOB$ is a right angle 1
 - iii) Describe the behavior of r as $x \rightarrow 0$ 1
- b) Prove by mathematical induction, for all odd positive integers n , $2^n + 7^n$ is divisible by 9. 4
- c) By equating the coefficients of $\sin x$ and $\cos x$, or otherwise, find constants A and B satisfying the identity
- i) $A(2\sin x + \cos x) + B(2\cos x - \sin x) \equiv \sin x + 8 \cos x$ 2
 - ii) Hence evaluate $\int \frac{\sin x + 8 \cos x}{2 \sin x + \cos x} dx$ 1

Question 14 continue over page

d)



Copy or trace the diagram into your writing booklet.

PCQ is the tangent at C; $PCQ \parallel MN$.

- | | | |
|-----|--|---|
| i) | Prove ABNM is a cyclic quadrilateral. | 2 |
| ii) | BM and AN, both produced meet the circle again at X and Y.
Prove $XY \parallel PCQ$. | 2 |

END OF PAPER

2012- EX1 Y12- Half Yearly- Solution

1. $27 - 2 + 3 - 2 + 9 = 35$

A

2. $\lim_{\theta \rightarrow 0} \frac{\tan \frac{\theta}{6}}{\theta} = \frac{1}{6} \lim_{\theta \rightarrow 0} \frac{\tan \frac{\theta}{6}}{\frac{\theta}{6}} = \frac{1}{6}$

B

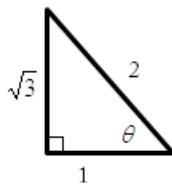
3. $x^2 - 2x + 2y - 11 = 0$
 $x^2 - 2x + 1 = -2y + 12$
 $(x - 1)^2 = -2(y - 6)$

$\Rightarrow a = \frac{1}{2}$

$\therefore S(1, 5\frac{1}{2})$

A

4.



D

5. $\lim_{x \rightarrow \infty} \frac{x^2 - 1}{x - 4} = \lim_{x \rightarrow \infty} \frac{\frac{x^2}{x} - \frac{1}{x}}{\frac{x}{x} - \frac{4}{x}} = x$

D

6. $\log_3 \frac{1}{81} = \log_3 3^{-4} = -4$

D

7. At A $f(x) = f'(x) = 0$

A

8.

C

$$9. \quad f(x) = \frac{x^2 - 2x - 3}{(x+1)(x-4)} = \frac{(\cancel{x+1})(x-3)}{(\cancel{x+1})(x-4)}$$

$f(x)$ has a vertical asymptote at $x = 4$

$f(x)$ is discontinuous at $x = -1$

C

$$10. \quad P(x) = x^3 + px^2 + qx + r$$

$$\begin{aligned} \sum \alpha &= \sqrt{\beta} - \sqrt{\beta} + \alpha \\ &= \alpha \\ &= -p \end{aligned} \tag{1}$$

$$\begin{aligned} \sum \alpha\beta &= \sqrt{\beta} \times -\sqrt{\beta} + \alpha\sqrt{\beta} - \alpha\sqrt{\beta} \\ &= -\beta \\ &= q \end{aligned} \tag{2}$$

$$\begin{aligned} \sum \alpha\beta\gamma &= -\sqrt{\beta} \cdot \sqrt{\beta} \cdot \alpha \\ &= -\alpha\beta \\ &= -r \end{aligned} \tag{3}$$

Sub. (1) & (2) into (3)

$$\Rightarrow \quad -p \times (-q) = r$$

$$\therefore \quad r = pq$$

D

Question 11

Mark

$$(a) \frac{1}{\sqrt{5}+1} + \frac{1}{\sqrt{5}+3} = \frac{\sqrt{5}-1}{4} - \frac{\sqrt{5}-3}{4}$$

$$= \frac{1}{2} \text{ is rational}$$

1

1

$$(b) y' = 2e^{2x} (\cos x - \sin x) + e^{2x} (-\sin x - \cos x)$$

$$= e^{2x} (2\cos x - 2\sin x - \cos x - \sin x)$$

$$= e^{2x} (\cos x - 3\sin x)$$

1

1

$$(c) \frac{x-1}{2x-1} \leq 3 \quad x \neq \frac{1}{2}$$

$$(x-1)(2x-1) \leq 3(2x-1)^2$$

$$(2x-1)(3(2x-1) - x + 1) \geq 0$$

$$(2x-1)(5x-2) \geq 0$$

1

1

$$\therefore x > \frac{1}{2} \quad \text{and} \quad x \leq \frac{2}{5}$$

1

$$(d) \frac{4^{n-1} \times 2^n}{8^{n-3}} = \frac{2^{2n-2} \times 2^n}{2^{3n-9}}$$

1

$$= 2^7$$

1

$$(e) y = 3x + 5, \quad y = mx + 4$$

$$\tan 45^\circ = \left| \frac{3-m}{1+3m} \right|$$

$$= 1$$

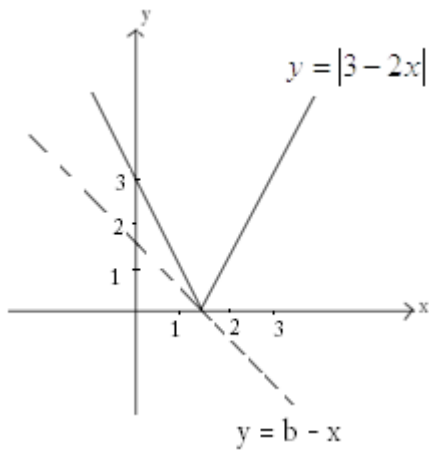
1

$$\frac{3-m}{1+3m} = 1 \quad \text{or} \quad \frac{3-m}{1+3m} = -1$$

$$\Rightarrow \begin{aligned} 3-m &= 1+3m \\ 4m &= 2 \\ m &= \frac{1}{2} \end{aligned} \quad \text{or} \quad \begin{aligned} 3-m &= -1-3m \\ 2m &= -4 \\ m &= -2 \end{aligned}$$

1 mark for each m

- 1 (for any mistake, incorrect scale, or no labelling of axes)



As the line $y = b - x$ moves up from the current position, it will cut the graph $y = |3 - 2x|$ at two points.

$$\therefore b > 1\frac{1}{2}$$

1

Question 12

$$\begin{aligned}
 \text{(a)} \quad \int \sqrt{x} \left(1 + \frac{1}{x}\right) dx &= \int x^{\frac{1}{2}} + x^{-\frac{1}{2}} dx \\
 &= \frac{2}{3} x^{\frac{3}{2}} + 2x^{\frac{1}{2}} + C
 \end{aligned}$$

1

$$\text{(b)} \quad \int_{\sqrt{2}}^2 \frac{4x+2}{x^2+x+1} dx = \left[2 \ln(x^2+x+1) \right]_{\sqrt{2}}^2$$

1

$$\begin{aligned}
 &= 2(\ln 7 - \ln(3 + \sqrt{2})) \\
 &= 2 \ln \left(\frac{7}{3 + \sqrt{2}} \right)
 \end{aligned}$$

1

(correct simplified answer)

$$\text{(c) i)} \quad f(x) = \sin x \cdot \tan^2 x$$

$$\begin{aligned}
 f(-x) &= \sin(-x) \cdot \tan^2(-x) && \mathbf{1 \text{ (substitution of } -x \text{)}} \\
 &= -\sin x \cdot (-\tan x)^2 \\
 &= -\sin x \cdot \tan^2 x \\
 &= -f(x) && \mathbf{1 \text{ (for correct following steps, no fudging)}}
 \end{aligned}$$

$\therefore f(x)$ is an odd function. **- 1 (for no conclusion)**

$$\text{Hence } \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \sin x \cdot \tan^2 x \, dx = 0$$

1

$$\begin{aligned}
 \text{(d)} \quad \int x(2x^2 + 1)^{\frac{5}{3}} dx & \qquad \text{let } u = 2x^2 + 1, \qquad du = 4x \, dx \\
 &= \frac{1}{4} \int u^{\frac{5}{3}} du && \mathbf{1 \text{ (correct changing variables)}}
 \end{aligned}$$

$$= \frac{1}{4} \times \frac{3}{8} \times u^{\frac{8}{3}} = \frac{3}{32} u^{\frac{8}{3}}$$

$$\therefore \int x(2x^2 + 1) dx = \frac{3}{32} (2x^2 + 1)^{\frac{8}{3}} + C$$

1 (correct answer)

Question 12 (cont.)

(e) $\frac{1 + \sin^2 x}{\cos^2 x} = 2 \sec^2 x - 1$

L.H.S = $\frac{1}{\cos^2 x} + \frac{\sin^2 x}{\cos^2 x}$ 1

= $\sec^2 x + \tan^2 x$

= $\sec^2 x + \sec^2 x - 1$

= $2 \sec^2 x - 1$ 1

$$\int_0^{\frac{\pi}{3}} \frac{1 + \sin^2 x}{\cos^2 x} dx = \int_0^{\frac{\pi}{3}} 2 \sec^2 x - 1 dx$$

= $[2 \tan x - x]_0^{\frac{\pi}{3}}$ 1 (correct integration)

= $2\sqrt{3} - \frac{\pi}{3}$ 1 (correct answer)

(f) At A, $\sin x = \cos x \Rightarrow x = \frac{\pi}{4}$

A = $\int_0^{\frac{\pi}{4}} \sin x dx + \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cos x dx$ 1

= $[-\cos x]_0^{\frac{\pi}{4}} + [\sin x]_{\frac{\pi}{4}}^{\frac{\pi}{2}}$ 1

= $2 - \frac{2}{\sqrt{2}}$ or $\frac{4 - \sqrt{2}}{2}$ 1

Question 13

(a) i) Domain : $x > 0$ 1

ii) $f'(x) = 1 + \frac{1}{x} > 0$ for all $x > 0$ 1

$\therefore f(x)$ is increasing for all x in the natural domain

iii) $f(0.5) = 0.5 + \log_e 0.5$
 $= -0.1935 < 0$ 1 (for both correct steps plus conclusion)

$$f(1) = 1 + \ln 1$$

$$= 1 > 0$$

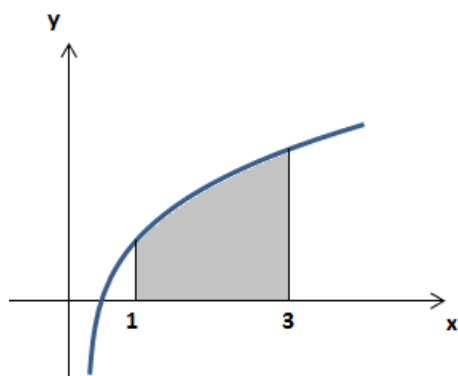
$\therefore$ The curve cuts the x -axis between $x = 0.5$ and $x = 1$

iv) $f'(0.5) = 1 + \frac{1}{0.5}$ 1
 $= 3$

$$a_1 = 0.5 - \frac{-0.1931}{3}$$

$$= 0.564$$
 1

(b)



$$y = \log_e 2x$$

$$\Rightarrow x = \frac{e^y}{2}$$

$$\therefore x^2 = \frac{1}{4} e^{2y}$$
 1 (correct changing variables)

$$V = \frac{\pi}{4} \int_1^3 e^{2y} dx = \frac{\pi}{4} \times \frac{1}{2} [e^{2y}]_1^3$$
 1

$$= \frac{\pi}{8} (e^6 - e^2) \text{ unit}^3$$
 1

Question 13 (cont.)

(c) i) $\ln\left(\frac{x-1}{x+1}\right) = \ln(x-1) - \ln(x+1)$

$$\frac{d}{dx} \ln\left(\frac{x-1}{x+1}\right) = \frac{d}{dx} [\ln(x-1) - \ln(x+1)] \quad \mathbf{1}$$

$$= \frac{1}{x-1} - \frac{1}{x+1}$$

$$= \frac{2}{(x-1)(x+1)} \quad \mathbf{1}$$

ii) $\int_2^5 \frac{2}{(x-1)(x+1)} dx = 2 \left[\ln\left(\frac{x-1}{x+1}\right) \right]_2^5$

$$= 2 \left(\ln\left(\frac{2}{3}\right) - \ln\left(\frac{1}{3}\right) \right) = 2 \ln 2 \quad \text{or} \quad \ln 4 \quad \mathbf{1}$$

(d) i) $y_1 = kx^n \Rightarrow y_1' = nkx^{n-1}$

$$y_2 = \log_e x \Rightarrow y_2' = \frac{1}{x}$$

Both functions have the same tangent at $x = a$.

$$\therefore m_1 = m_2 \quad \text{i.e.} \quad nkx^{n-1} = \frac{1}{x}, \quad \text{Sub. } x = a \quad \mathbf{1 \text{ (showing } m_1 = m_2)}$$

$$\Rightarrow nka^{n-1} = \frac{1}{a} \quad \therefore \quad a^n = \frac{1}{nk} \quad (1) \quad \mathbf{1}$$

$$\text{also} \quad a = \frac{1}{\sqrt[n]{nk}} \quad (2)$$

ii) y_1 and y_2 have the point of intersection at $x = a$

$$\therefore kx^n = \log_e x$$

$$\text{At } x = a, \quad ka^n = \log_e a \quad \text{Sub. (1) \& (2)} \quad \mathbf{1 \text{ (showing } y_1 = y_2 \text{ at } x = a)}$$

$$\Rightarrow k \times \frac{1}{nk} = \log_e(nk)^{-\frac{1}{n}}$$

$$\frac{1}{n} = -\frac{1}{n} \log_e(nk)$$

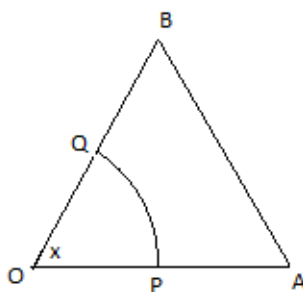
$$\Rightarrow e^{-1} = nk$$

$$\therefore k = \frac{1}{en}$$

1

Question 14

(a)



i)

$$A_{\triangle OAB} = \frac{1}{2} a^2 \sin x,$$

$$A_{\text{sector } OPQ} = \frac{1}{2} r^2 x$$

1 (correct finding both areas)

$$\text{Given } A_{\triangle OAB} = 2 A_{\text{sector } OPQ}$$

$$\therefore \frac{1}{2} r^2 \sin x = r^2 x$$

$$\Rightarrow r^2 = \frac{a^2 \sin x}{2x}$$

1

$$\text{ii) when } x = \frac{\pi}{2},$$

$$r^2 = \frac{a^2}{\pi}$$

$$\Rightarrow r = \frac{a}{\sqrt{\pi}}$$

1

iii)

$$\lim_{x \rightarrow 0} \frac{a^2 \sin x}{2x} = \frac{a^2}{2} \lim_{x \rightarrow 0} \frac{\sin x}{x}$$

$$= \frac{a^2}{2}$$

1

(-1 for missing or incorrect conclusion)

- (b) i) Prove it is true for $n = 1$ 1
 $2^1 + 7^1 = 9$ is divisible by 9
 $\therefore$ it is true for $n = 1$.

Assume it is true for $n = k$
i.e. $2^k + 7^k = 9P$, where P is an integer.

$$\Rightarrow 7^k = 9P - 2^k \quad (1)$$

Prove it is true for $n = k + 2$ (odd integer).

i.e. to prove: $2^{k+2} + 7^{k+2} = 9Q$, where Q is an integer. 1

$$\begin{aligned} \text{L.H.S} &= 4 \times 2^k + 49 \times 7^k \\ &= 4 \times 2^k + 49 \times (9P - 2^k) && \text{Using (1)} \\ &= 9 \times 49P + 4 \times 2^k - 49 \times 2^k \\ &= 9 \times 49P + 45 \times 2^k \\ &= 9 (49P + 5 \times 2^k) \\ &= 9Q && \text{where } Q = 49P + 5 \times 2^k \end{aligned} \quad \begin{matrix} 1 \\ 1 \\ 1 \end{matrix}$$

$\therefore$ It is true for $n = k + 2$.

The formula is true for $n = 1$, true for $n = 2$ (from step 3), $n = 3$ and so on for any odd integer. Hence by mathematical induction $2^n + 7^n$ is divisible by 9 for all odd positive integers.

-
- (c) i) $A(2\sin x + \cos x) + B(2\cos x - \sin x) \equiv \sin x + 8\cos x$.

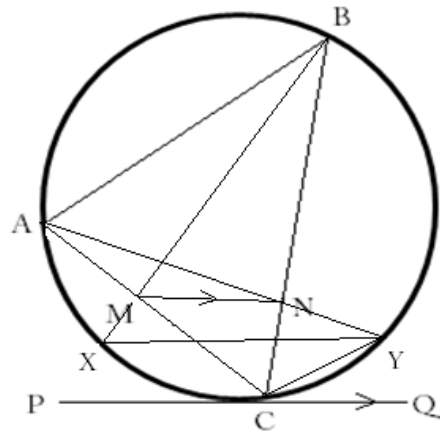
$$\begin{aligned} \text{L.H.S} &= 2A\sin x + A\cos x + 2B\cos x - B\sin x \\ &= (A + 2B)\cos x + (2A - B)\sin x \\ &\equiv \sin x + 8\cos x. \end{aligned}$$

$$\begin{aligned} \therefore \quad A + 2B &= 8 \\ 2A - B &= 1 \quad (\times 2) && \Rightarrow \quad \begin{array}{r} 4A - 2B = 2 \\ + \\ A + 2B = 8 \\ \hline 5A = 10 \end{array} \end{aligned}$$

$$\therefore \quad A = 2 \text{ and } B = 3 \quad \text{1}$$

$$\begin{aligned} \text{ii) } \int \frac{\sin x + 8\cos x}{2\sin x + \cos x} dx &= \int \frac{2(2\sin x + \cos x) + 3(2\cos x - \sin x)}{2\sin x + \cos x} dx \\ &= 2 \int dx + 3 \int \frac{2\cos x - \sin x}{2\sin x + \cos x} dx && 1 \\ &= 2x + 3 \ln(2\sin x + \cos x) + C && 1 \end{aligned}$$

(-1 for missing reason)



$\angle ABC = \angle ACP$ (angles in the alternate segment)

$\angle NMC = \angle ACP$ (alternate angles, $MN \parallel PC$)

$\therefore \angle ABC = \angle NMC$

1

$\angle AMN + \angle NMC = 180^\circ$ (supplementary angles)

$\therefore \angle ABC + \angle AMN = 180^\circ$

$\therefore$ ABNM is a cyclic quadrilateral (sum of interior opposite angles = 180°)

1

Hence $\angle MAN = \angle MBN$ (angles in the same segment)

1

Also $\angle MBN = \angle XYZ$ (angles in the same segment)

and $\angle MAN = \angle YCQ$ (angle in the alternate segment)

$\therefore \angle XYZ = \angle YCQ$

1

Hence $XY \parallel PCQ$ (alternate angles are equal in parallel lines).